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STUDY OF VARIOUS MAINTENANCE APPROACHES TYPES OF FAILURE AND FAILURE DETECTION TECHNIQUES USED IN HYDRAULIC PUMPS: A REVIEW

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Abstract

Pumps are considered to be one of the most important elements in many industrial sectors, plants and also in the manufacturing units. The pumps are the key elements in the agriculture, food industry, oil industries, gas industries, water treatment plants, paper industries, sewage, chemical plants, furnaces, reactors, heat exchangers, hydraulic power service, mines, pulp industry and so on. As the use of the pump is on the massive scale, there are few problems that will occur in the pumps leads to failure. Unexpected failure of the pumps become more costly due to the interruption in process and further cost are incurred in terms of replacement and repair. Failure and unscheduled breakdown in the pumps leads to the loss of the production and subsequently loss in the revenue in the industries and production units. Over the recent years, the significance of maintenance and the maintenance management is grown up within the manufacturing organizations. Maintenance engineering helps a lot in reducing cost, improve productivity and increase profitability for the manufacturing units throughout the world. Fault prognosis and diagnosis have an effective role for the safe operation and long life of the working system. In this paper, we studied about the type of failure in pumps and the best suited maintenance approach and various failure early detection techniques. This paper will explain and review the need of maintenance approach along with early detection of incipient failure which helps to make the right decision so that to increase the lifetime of the system, efficiency of the system by reducing cost.

Keywords: - Pumps, Maintenance, Prognosis, Diagnosis, Failure.

1.0 INTRODUCTION:

The maturation in the world and the world's economy is the major challenge to the production, market, and industries which leads to modification up to next level [1]. One-fifth of the total energy consumption in the industry is caused by the hydraulic systems [2]. The most common machine used for the transfer of liquid is pumps. Pump can be defined as the mechanical device that imparts energy to the fluid. They make use of mechanical energy, which usually come from electric motor to increase the pressure and kinetic energy of the fluid [3]. The pumps are the key elements in the agriculture, food industry, oil industries, gas industries, water treatment plants, paper industries, sewage, chemical plants, furnaces, reactors, heat exchangers, hydraulic power service, mines, pulp industry and so on [4]. Pumps are considered to be one of the most prominent machinery used in many industrial sectors, plants and also in the manufacturing units. Due to the large scale usage of pumps in the manufacturing world, it is important to study the various types of failures, various failure detection techniques and the maintenance approaches that are being employed so as to prevent the incipient failure in the pumps. To monitor the condition of the pumps systems, a great deal efforts has been made to investigate, detect and diagnose incipient faults and failure in the pumps through the use of fault diagnosis tools [5].

2.0 PUMPBASICS AND CRITICAL COMPONENTS:

Pumps come in a variety of sizes that can be used for a wide range of applications. These pumps can be classified according to their basic principle of operation i.e. displacement and dynamic pump. Displacement pumps can be further classified as rotary pumps and reciprocating pumps. Dynamic pumps can be classified as centrifugal pumps and special effect pumps. Rotary pump are the positive displacements pump and in this type of pump for each revolution of the pump, a fixed volume of fluid is moved regardless of the resistance against which the pump is pushing. It consists of the vanes, gears, lobes, or screws to trap and convey fluid from the inlet to the outlet of the pump. Reciprocating pump is another class of positive-displacement pumps which includes the piston pump, plunger pump and diaphragm pump [6] .This type of Pump use the back and forth motion of mechanical parts, such as piston, diaphragm, to pressurize the fluid. It is often used where a relatively small quantity of liquid is to be handled and where delivery pressure is quite large [7]. In these pumps, the chamber in which the liquid is trapped is a stationary cylinder that contains the piston or plunger. Windmill water pump, axial piston pump, Radial piston pump and oil pump are some of the examples of the reciprocating pumps [8]. Centrifugal pumps use the centrifugal force imparting to the fluid by the rotating elements called impeller to increase the kinetic and pressure energy of fluid. A centrifugal pump is a rotating machine that comprises of six main parts. These parts works together to keep the pumps operation properly. These components include an impeller, a bearing, a bearing frame, a pump casing, a mechanical seal and a shaft.[9] These all are the components that directly affect the desired pump characteristics. The two main parts are the impeller and the pump casing [10]. Impeller is the moving part which is attached to shaft and driven by the motor [11]. It provides the centrifugal acceleration to the fluid. Pump casing is the stationary part in the centrifugal pump. The pump casing helps to houses the impeller capture and directs the water off the impeller. Fluid enters the eye of the impeller and exits the impeller with the help of centrifugal force [12]. The pressure developed by the pump will is directly in the relationship to the impeller diameter, the size of the impeller eye, the number of the impeller and the shaft speed.

There are some of the auxiliary components in the hydraulic pump such as seal flushing, quenching system, seal drains and vents bearing lubrication system, seal chamber and pump pedestal cooling system.

3.0 GENERAL PROBLEMS IN PUMPS:

In a hydraulic pump impeller, bearing and seal are the critical components that directly affect the desired pump characteristics.

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In the hydraulic pumps these are some of the serious problems that lead to the defect in the pumps. These are:-

- Cavitation
- High vibration
- Misalignment
- Defect on the impeller
- Defect in the seal and defective bearing

Cavitation is a term that come from a Latin word "cavus", which means cavity or a hallow space. Cavitation may also be defined as the rapid formation and collapse of the bubble that create cavities in a very low pressure area [13]. Some other terms such as vapour pocket, bubbles, gas pocket are used in place of the cavity. These bubbles are formed where the vapour pressure of the liquid and saturation pressure of the gas is more than that of the local static pressure [14]. In most of the cases this phenomenon of cavitation is due to the insufficiency of the available Net Positive Suction Head (NPSH). This problem leads to degradation of efficiency i.e. reduced pump performance, excessive vibrations, cavitation erosion, excessive noise, and wear the pump part. The main cause of gaseous cavitation is loss of capacity [15]. FIGURE 3.1 representing cavitation defect in impeller.

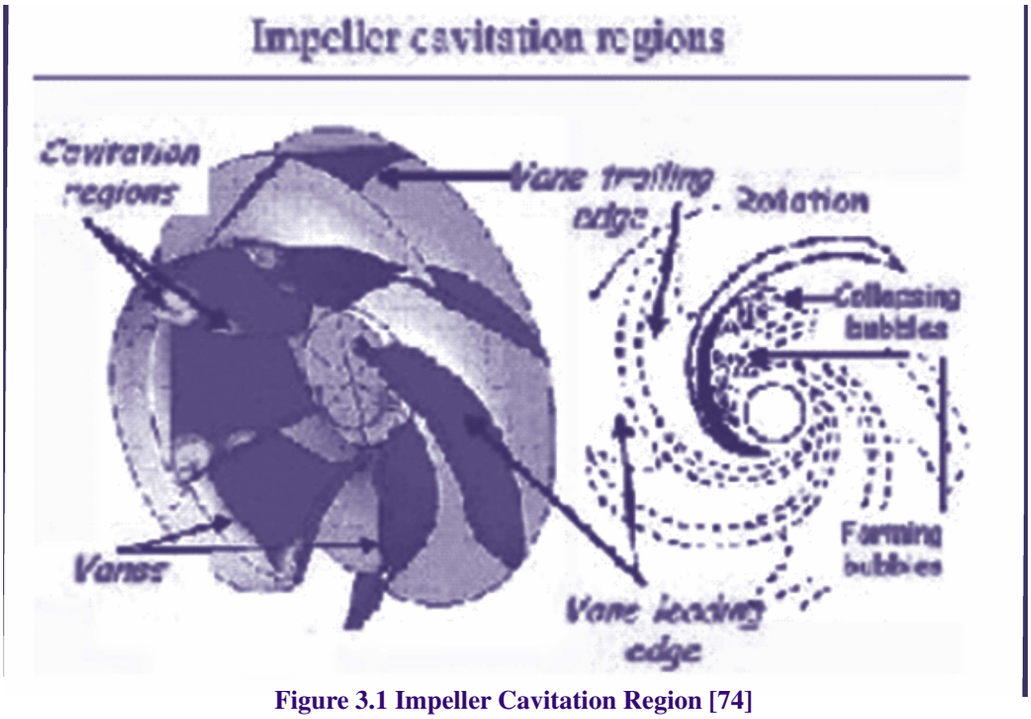

Figure 3.1 Impeller Cavitation Region [74]

Another major problem in the pump is high vibration which leads to the misalignment in the parts of the system [16]. Improper installation or maintenance, incorrect application, hydraulic interaction with the piping system, or design and manufacturing flaws are some of the reasons that lead to problem in the pumps [17]. Seal breakage leads to the leakage in the system which affects the performance of the system [18]. It is one of the most common problems in the centrifugal pumps [19]. Seal and bearing defect cause number of the serious problems which leads to leakage abnormal noise, high vibrations, loss of flushing, loss of lubrication, contamination of oil, and leakage from pump casing problem [20]. Some other problems in the centrifugal pumps are overheating and suction clog. FIGURE 3.2 representing broken impeller with fault seal.

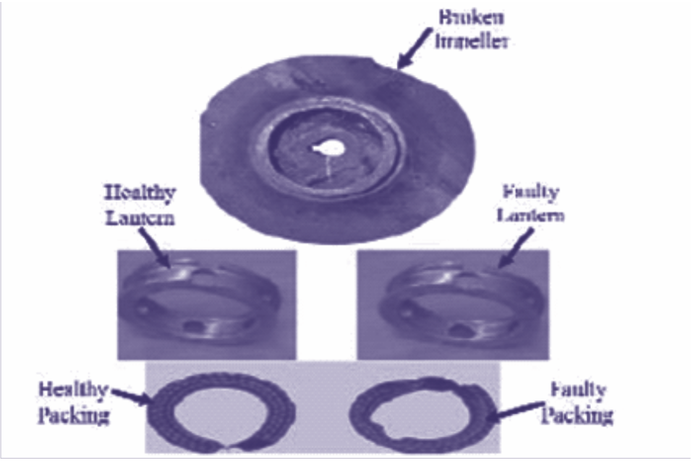

Figure 3.2 Broken Impeller with Fault Seal [75]

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4.0 MAINTENANCE AND ITS IMPORTANCE:

Maintenance can be defined as the combination of both technical and administrative actions that are intended and performed to retain and restore a system. The main objectives of the maintenance system are to ensure system proper functioning, ensuring the system life, ensuring the safety and ensuring the human well-being [21]. Maintenance planning which is based on the risk analysis helps in minimizing the probability of the system failure. Maintenance techniques play a very vital and important role in the mechanical engineering and its applications.

4.1 MAINTENANCE TYPE:

- Reactive Maintenance (RM) This type of maintenance technique is also called as (Run-to-Failure) breakdown maintenance. It takes action on the equipment only when the defect appears in the system [22]. The attention is paid only on the defect but no attention is paid over the root cause of the failure. Reactive maintenance is characterized by two elements i.e. bad, poor planning and patchy repair. It is a traditional technique of maintenance and is not used by most of the manufacturing organizations. Diagnosis is done in the system after the failure happens [23].
- Preventive Maintenance (PM) It only emphasizes on the prevention but no attention is paid toward the root main cause of failure removal from the system [24]. This maintenance policy helps us and tries to determine the replacement, series of check, or component revision with the frequency related to failure rate [25]. It is a maintenance strategy which is time driven and it helps to schedules maintenance for a machine and its component which is based on the experience of the mean time between failures (MTBF) and mean time to repair (MTTR) [26].
- Predictive Maintenance (PdM) This type of maintenance activity uses specific instruments such as wear analyzers, vibration analyzers; temperature analyzers that ensure the maintenance intervention might be executed and is carried out before the appearance of the defect [27]. It also helps in improving the availability of the process equipment in increasing the life time, safety, quality and reducing the maintenance cost of the system. These diagnostic techniques and the essential tool for a predictive maintenance program [28]. By evaluating the technical operating condition basically by symptoms such as noise, vibration, temperature variation and pressure variation the maintenance staff will able to predict the failure in the system [29].
- Condition Based Maintenance (CBM) In this type of the maintenance activity a continuous or periodic survey over the general working condition of the system take place which clearly point out the abnormal situation and symptoms of failure [30]. CBM technique provides input for the prognostic models in PHM and support the timely maintenance and accurate decision making that prevents downtime and increase the availability of the system [31]. CBM consist of three basic steps: - (1) Data Acquisition, which help to obtain relevant system health parameters. (2) Data Processing, it is the second step which helps to analyze and interpret the data which is collected in the step first for the better understanding of the input data. (3) Maintenance Decision making, is the final step which helps in recommending the effective maintenance policy which is based on the data from second step [32].
- Reliability Centered Maintenance (RCM) is most suitable and effective for the uncertainty laded system, where the behavior of the system cannot be predicted accurately based on the previously made observation and the domain knowledge [33]. RCM not only enables us to create a maintenance strategy to address the main cause of failure but also help to provide the systematic approach that will define the routine maintenance program [34]. This type of maintenance approach mainly focus on the ability of the system to have expected reliability up to a certain period of time and utilize some of the statistical tools such as Failure mode effect analysis (FMEA), Failure mode effect critical analysis (FMECA), Fault tree approach (FTA) to retrieve the information that can help to identify failure [35]. This maintenance approach helps to reduce the unnecessary, untimely and unscheduled maintenance [36]. *Risk Based Maintenance (RBM)* In this type of maintenance approach we first consider the level of the risk of failure in the system. After determining the risk it is been evaluated and then the maintenance is been done in the system [37]. The three main module of RBM are :- (1) Risk determination, which is the basic step and it involve risk identification and risk estimation (2) risk evaluation, which is the second step and consist of risk aversion and risk acceptance analysis (3) maintenance planning considering risk factor is the final and last step in this Risk based maintenance approach [38].
- Total Productive Maintenance (TPM) This activity is implemented in many of the manufacturing and production sectors to improve their equipment efficiency and to obtain the competitive advantage in the global market in terms of both cost and quality of the product. It is considered to be an improvement program which establishes a comprehensive productive-maintenance system throughout the entire life of the equipment [39].The Total Productive Maintenance implementing process has been fraught with pitfalls which make it a difficult task. The pitfalls in the total productive maintenance are the lack of management support, lack of involvement of production associates, lack of resources and lack of sustained momentum [40].

Software Based Maintenance Use of the computer based technology and software have brought a drastic change in the manufacturing systems, system maintenance management that include the planning, scheduling, documentation of the history, predicting the system reliability, maintainability, controlling and also help in the decision making [41]. There are some software packages for the maintenance optimization which are: - REGINA, FRANTIC, MAINOPT, AGE/CON, PERDEC, RELCODE, KMOSS, OPTIMON all include analytical model. Computerized maintenance management system (CMMS) is also an important tool which is used for the maintenance management in many organizations worldwide. In CMMS, the data collector agents are assigned to the critical parts, these data collector agents are activated to work on the monitoring. Then the data collected is summarized statically and then results are compared with the Weibull distribution [42]. This system proposed and suggests the maintenance engineers to make a prediction on the reliability of the system, failure probability and the remaining life in the future. FIGURE 4.2 representing maintenance transformation map.

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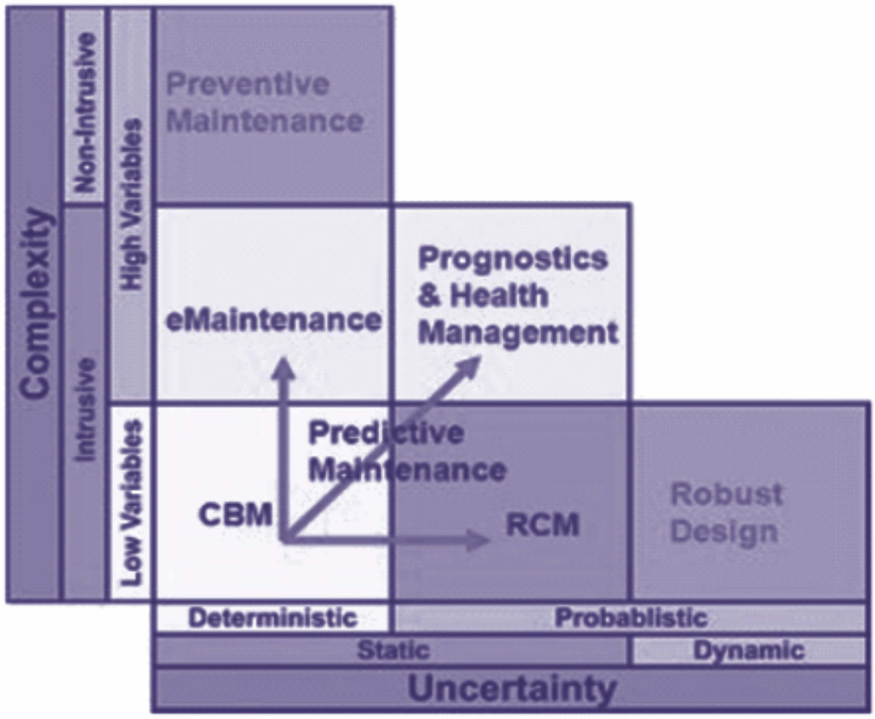

Figure 4.2Maintenance Transformation Map [76]

5.0 PROGNOSIS AND DIAGNOSIS:

The prognostics and health management (PHM) is a critical research domain which act as the advanced predictive tools [43]. It not only helps to determine the equipment performance but also help to estimate the time of the failure and reduce the impacts of these uncertain failures [44]. Many researchers have been conducted in PHM and are considered as an emerging field in the mechanical engineering. Most of the PHM applications addresses forecasting and helps to predict the remaining useful life (RUL) of that system. It also aid in making the modern maintenance strategy [45]. Prognosis and diagnosis are different from each other in the conceptual way. The word prognosis comes from a Greek word which means fore-knowing and fore-sensing [46]. This type of technique is new in the engineering field and it help to predict the fault before the failure happens. On the other hand diagnosis is the identification of the nature and cause of the certain phenomenon. This type of technique is used after the failure of the system to detect the root cause of that failure [47]. The main purpose of the prognosis is to predict the event before its possible occurrence, so in this time is an important variable which is not in case of the diagnosis. FIGURE 5.1 represents the difference perception of diagnostics and prognostics.

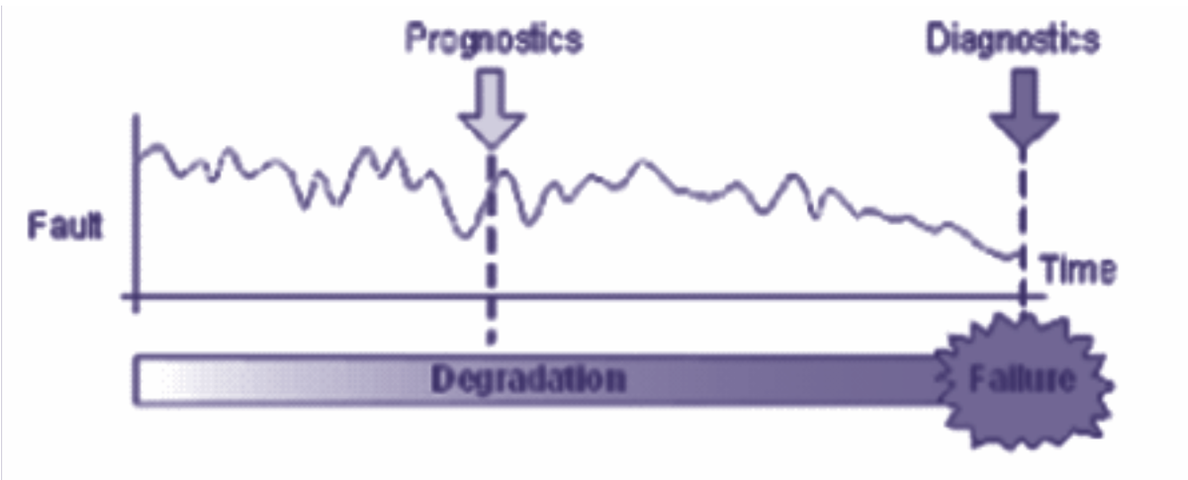

Figure 5.1 Differencing Perceptions Of Diagnostics And Prognostics.[76]

The concept and framework of the prognosis have been developed from the well-known maintenance strategy and diagnostic technique i.e. Preventive maintenance, Condition based maintenance, reliability centered maintenance [48]. FIGURE 5.2 representing overview of condition monitoring and maintenance processes.

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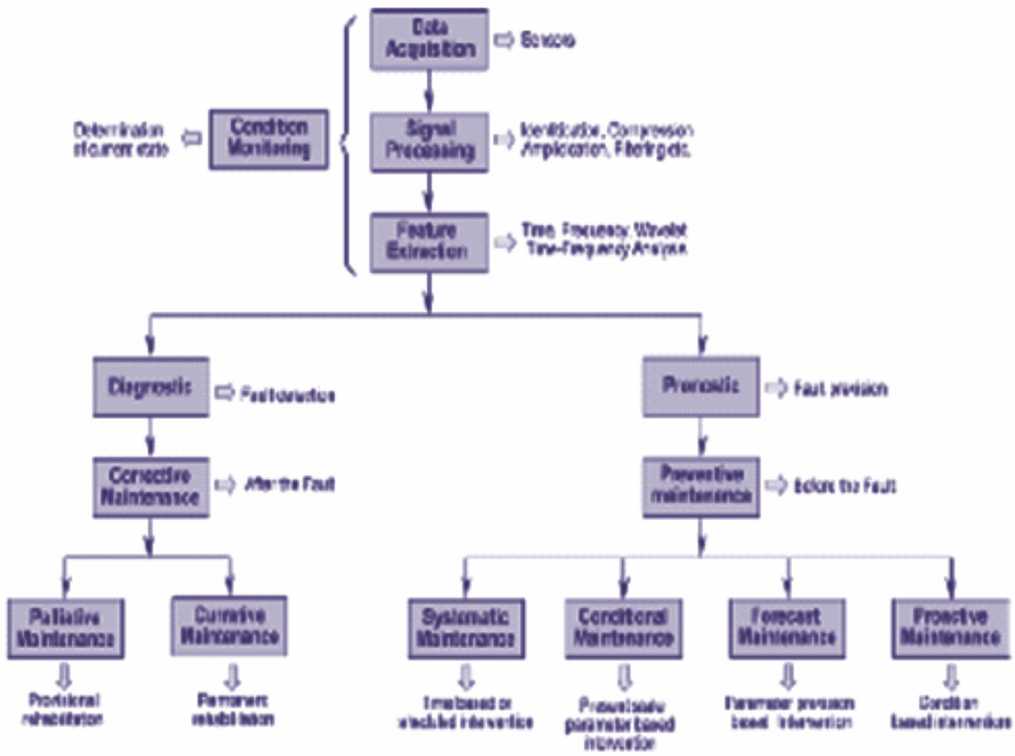

Figure 5.2 representing overview of condition monitoring and maintenance processes. [77]

6.0 FAILURE EARLY DETECTION TECHNIQUES:

Vibration analysis, oil analysis, temperature analysis, acoustic emissions, ultrasonic systems are some of the techniques which are generally employed for the prognosis of the system for a long period of time. Because of the excellent capabilities in describing the system performance these techniques are used and widely employed. Sensor fusion technique is generally used to interpret information from the multiple servers [49]. Prognostics approach is sub categorized in to three classes i.e. model based, data driven, and hybrid prognosis approach. In the model based approach the data is generated from the simulated model under the degradation and normal condition and then this data is simulated under multiple operational modes and the extracted data weighted with time-averaged probability model. In the data-driven prognostic approach we can determine the remaining useful life by determining the trajectory of the developing fault and predict the time when it will reach to the threshold level. Alpha-Beta-Gamma and Kalman filter are the two main prediction and tracking tool that are used in this model. They can track the changes in the vibration level, mode frequency, and other waveform feature and can determine the future failure hazard, remaining useful life (RUL) and the probability of the survival [50]. In the hybrid approach both these Model based and Data-Driven model are fused and the prognosis is been done. It is compared to be more reliable and accurate than that of the Model based and Data-Driven approach. Some of the commonly used algorithms used are Neural Network (NN), Fourier transformation (FT), Short Time Frequency Transform (STFT), Support Vector Machine (SVM), Fuzzy Logic, Neuro-Fuzzy Hybrid Model. But, currently development in the Neural Network has added new dimensions which overcome the existing problem of the prognosis and diagnosis [51]. Analysis of vibration, acoustic emission, frequency analysis, band width analysis acoustic sound, pressure analysis, flow rate analysis and temperature analysis are some of the commonly used monitoring techniques. Vibration analysis and acoustic emission are widely used in the condition monitoring of the rotating machines [52]. Vibration analysis has been used in the fault diagnosis for decade by measuring and analyzing the vibration of the machine. These techniques help us to determine the nature and severity of the defect and hence predict the machine's life or failure point [53]. Usually, accelerometer mounted on the outer surface of the bearing housing is used to acquire the vibration signals [54]. These signals consist of the vibration from the meshing gears, bearing, shaft and other components of the pump and then the vibration signals are captured in different frequency areas [55]. The level of the vibration can be compared with the historical baseline value to determine the severity. By examine these frequencies the analysts can identify the type of the problem and its root cause [56]. Another widely used approach for the diagnosis of the fault is Acoustic Emission Technology. Acoustic emissions are defined as the transient elastic waves generated from a rapid release of the strain energy caused by a deformation within or on the surface of a material [57]. These emissions propagate on the surface of the material and the displacement of these waves is measured by using Acoustic Emission Sensors. This technology provides powerful diagnostic capabilities, which are efficient, cost-effective and safe [58]. AE Sensors are place on the non-rotating members of the machine, such as bearing or casing. The commonly used parameters used in AE for diagnosis are amplitude, crest factor, counts and events, R.M.S and energy [59]. Table 6.1 represents the introductory summarization of PHM tools of critical component

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Component	Pump
Issue & Failure	Defective Bearing, High Vibration Cavitation, Misalignment, Seal Breakage, Leakage
Common measure	Vibration Pressure Analysis, Acoustic Emission
Common features	Vibration Characteristics, Frequency Pressure Time Domain Statistical Characteristics.
Common algorithm	Neural Network(NN), Fourier Transformation(FT), Support Vector Machine (SVM), Fuzzy Logic(FL), Neuro -Fuzzy Hybrid Model, Rough Set, Principle Component Analysis (PCA)

Table.6.1.Introductory Summarization Of Phm Tools Of Pump

7.0 BACKGROUND RESEARCH:

The purpose to review the background research was to identify and analyze documents containing information relevant to pumps, their failures and measures so that we can improve the total life of the pump. It offers a framework from which to understand the basic concepts behind the operation of the pump and the reasons due to which failure may happens. **Muralidharan, Sugumaran & Indira. (2014) [60]** reported that the performance of bearing and impeller has direct effect on the desired pump characteristics. The faults considered in this study are bearing fault (BF), impeller fault (IF), bearing and impeller fault (BFIF) together and cavitation (CAV). SVM algorithm is used for practical applications of fault diagnosis of monoblock centrifugal pump. **Lin et al (2011, May) [61]** proposed some major failure that will come in the pumps while the operation. These are mechanical failure of bearing, seal and impeller that may lead to Erosion, corrosion, excessive noise, overheating, leakage, suction and discharge recirculation, over-heating etc. **Lebeck (1991) [62]** purposed that proper functioning of the mechanical seals depends on the lubrication system. **Wang & Yamaguchi. (2002) [63]** reported that Contamination failure is another major reason for the failure in the mechanical seals. Contamination failure includes Hang-up, clogging, contamination due to particles deposit and sticking. **Farokhzad & Ahmadi. (2013) [64]** used a neural network simulator which is built for prediction of cavitation in centrifugal pump. **Farokhzad, Ahmadi, Jaefari, Abad & Kohan. (2012) [65]** presents the design and development of artificial neural network based model for the fault detection of centrifugal water pump. **Sanz, Perera, & Huerta. (2007) [66]** presented a technique for monitoring the condition of rotating machinery from vibration analysis. **Wang & Chen. (2007) [67]** proposed the synthetic detection index with fuzzy neural network to evaluate the sensitivity of non-.dimensional symptom parameters for detecting faults in centrifugal pump. **Farokhzad, Ahmadi, Jaefari, Abad, & Kohan. (2012) [68]** investigated that the fault detection and diagnosis have an effective role for the safe operation and long life of systems. In order to effectively diagnose faults, they purposed an intelligent diagnosis method with the parameters in frequency domain based on the fast Fourier transform (fft). **Salimi & Niromandfam. (2015) [69]** studied the vibration situation of hydraulic pump and the defects were classified using nervous net. **Kankar, Sharma, & Harsha. (2011) [70]** investigated the use of artificial neural networks (ANN) for condition monitoring of the rotating machinery from the vibration time series. **Kamiel. (2015) [71]** investigated the use of a set of statistical features (e.g. Standard deviation, standard error, kurtosis, variance, range, skewness, sum minimum value and maximum value) for fault diagnosis of centrifugal pumps. **Mckee et al. (2015) [72]** used two statistical features (kurtosis and crest factor) and neural networks to detect cavitation in centrifugal pumps. **McKee, Forbes, Mazhar, Entwistle, & Howard. (2011, May) [73]** investigated about the major fault modes that are found in centrifugal pumps, especially those in the water and sewage industry. The problems that will be addressed here will be hydraulic failures (cavitation, pressure pulsations, radial thrust, axial thrust, suction and discharge recirculation), mechanical failures (bearing failure, seal failure, lubrication, excessive vibrations, fatigue), and other types of failure (erosion, corrosion, excessive power consumption).

8.0 CONCLUSION:

This paper presents a discussion on the various types of failures in the hydraulic pumps, role of maintenance policies along with early failure detection techniques. Maintenance is aimed to increase the availability of any system taking account of the environment safety and optimizing the total life cycle cost. So it is ensured that by developing and implementing the maintenance monitoring, prognosis and diagnosis techniques with in the maintenance system we can increase the lifetime of the system, reliability of the system, increase the uptime and hence reduce the maintenance cost. This discussion is based on the analysis of the different researchers regarding the prognosis, diagnosis, maintenance and the type of failures that will occur in the system. This will help in right decision making by the use of various mathematical model that will help to achieve effective and efficient maintenance.

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